California Median Housing Price Estimation via Linear Regression Modeling

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Abstract

This project aims to predict housing value based on a myriad of home buying factors. Incorporating such factors in home buying is crucial to the accuracy of home buying prediction. Utilizing the California Census Dataset published by the United States Census Bureau which includes metrics such as population, median income, median housing prices, and so on, this project aims to integrate these metrics into a comprehensive housing price estimation. Using predictive modeling techniques, the machine analyzed median income, longitude and latitude, and median house age to forecast affordability trends. To minimize error, data dropping and z-score based data removal was employed to reduce the amount of outliers present in the data. The results of the project indicate a strong positive correlation between predicted housing value and associated actual housing value, suggesting that such values introduced above did have a significant impact on predicting median house value.

1. Introduction

Housing affordability has become a pressing issue in California, affecting both new and returning homebuyers. While existing homebuyers must navigate budget constraints and geographical considerations, new homebuyers contend with these issues as well as the added difficulty of entering the rapidly moving market. Therefore, it is invaluable to both parties to consider the affordability of housing. However, it is difficult to predict the pricing of a specific unit while considering numerous factors such as population, median income, longitude and latitude, etc. This project attempts to incorporate these factors into a comprehensive linear regression model to provide a high accuracy range of pricing range a unit could fall under. By doing so, it offers a practical tool for prospective buyers and policymakers to better understand and respond to housing market dynamics

2. Data Preprocessing

For achieving the purpose of the minimum amount of error occurring in the linear regression model as possible, a sufficient amount of data cleaning was performed accordingly.

2.1 Ocean Proximity Data Removal

The removal of all string values from columns used in prediction, namely ocean proximity, was required for the successful run-time completion of the data. Although ocean proximity could be a housing value determinants, the nature of the column as a string limited its ability to be included in the regression model. Employing **pandas** library techniques, the entire ocean proximity column was removed[`df_clean = df_clean.drop 'ocean_proximity', axis=1)`]

2.2 Null Value Removal

The original dataset included many null data points as would be expected of a real world data set where not all factors as given or could be collected. However, these null points prove to be problematic to the overall regression model. Null values can skew data, bias results, and even break algorithms. By utilizing the **pandas** library used for data manipulation, null values were successfully dropped [`df_clean.dropna()`], greatly improving data accuracy.

2.3 Z-Score Based Data Removal

Outliers are extreme values that differ greatly from the mean data point. Including outliers, the dataset can become skewed and less representative. By removing data point two standard deviations away, the dataset was much more comprehensive and allowed the model to learn a more general pattern without being influenced by anomalies. Z-Score based data removal was used again with the **pandas** library first by identifying the individual z scores of all data points[`z_score = np.abs((df_clean['median_house_value'] - df_clean['median_house_value'].mean()) / df_clean['median_house_value'].std())`], then by keeping only data points within the two standard deviation range[`df_clean = df_clean[z_score < 2]`]

3. Feature Selection

Domain reasoning was used to determine that median income, longitude and latitude, and median housing age were the most logically important to deciding a property's value

4. Machine Learning via Linear Regression

4.1 Variable Initialization

The linear regression model requires a sufficient amount of factors to be included to predict a housing value. In this model, median income, longitude, latitude, and median housing age[`X = df_clean[["median_income", "longitude" ,"latitude", "housing_median_age"]]`] were used to predict median housing value[`y = df_clean[[ "median_house_value"]]`]. The amount of factors is critical to the user's experience by introducing more variables for a more accurate estimation.

4.2 Training and Test Data Initialization and Model Implementation

To predict values, the machine must have a set of data to train and a separate set of data to evaluate the model performance. The Python library **scikit-learn** provides the means to do so[`X_train, X_test, y_train, y_test = train_test_split (X,y,test_size=.2, random_state=42)`]. **Scikit-learn** separates the dataset into a 80-20 set where 80% of the data is used to train the model and 20% of the data is used to evaluate the model performance. This training data is then used via linear regression to fit the data[`model = LinearRegression()`] [`model.fit(X_train, y_train)`]. Once the data is fitted, the model can then make housing value predictions via the testing dataset[`y_pred = model.predict(X_test)`]. Even in the resulting data there are outliers that reduce the accuracy of the result. To remedy this problem, the maximum prediction value was set to $400,000[`y_pred = np.minimum( y_pred, 400000)`]. Additionally, the linear regression model occasionally predicts a negative housing

value which is not possible. To address this issue, the **numpy** Python library was put in use to set all negative predictions to zero[`y_pred = np.maximum(y_pred,0)`].

5. Graphical Interpretation

The Python library **matplotlib** was used to plot the predicted versus actual prices in a scatter plot. The results demonstrate a strong positive correlation between predicted and actual prices. [`plt.scatter(y_pred, y_test)`]

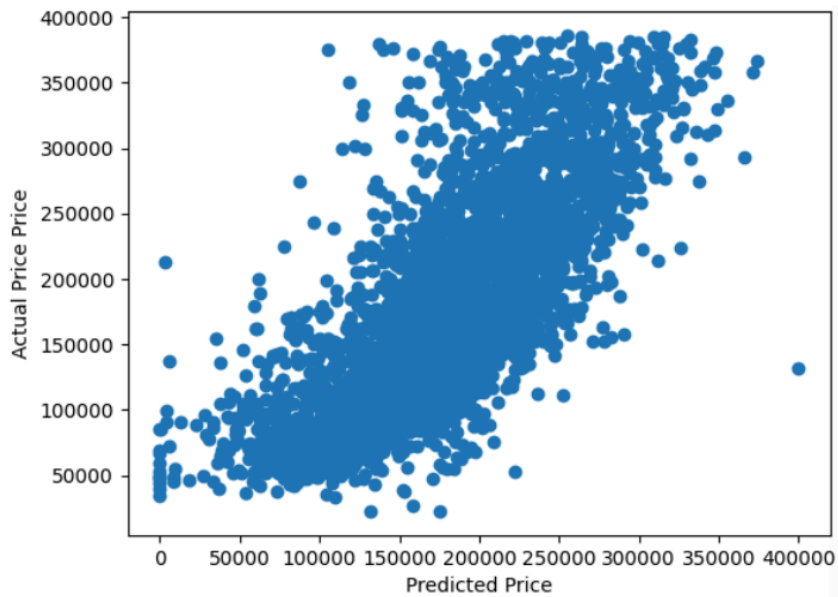

Figure 1. Predicted Versus Actual Housing Price via CoLab

6. Error Analysis and Interpretation

6.1 R^2 Value

The computed R^2 value[`r_squared = f"{r2:.2f}"`] for prediction was 0.57. Although this value seems relatively low for a predictive model, it reflects the inherent variability in housing prices due to numerous unobserved and incalculable variables such as local amenities, neighborhood characteristics, market fluctuations, and seller sentiment.

6.2 R Value

This model computed an R value of 0.75 suggesting a strong positive correlation between the predicted and actual price.

6.3 Error Interpretation

Despite the relatively low R^2 value obtained for the model, its predictions should not be interpreted as exact figures. Rather, the model is best understood providing **high-accuracy estimate ranges** for the potential prices of an inputted property. This approach acknowledges the

variability housing prices face that are influenced by numerous non-numerical factors. In practice, this aligns with how commercial housing price predictors operate: presenting buyers with an accurate **range** or **confidence interval** rather than a definite price point. Providing a range of values rather than exact values allows the model to offer a more realistic insight for first-time buyers and existing homeowners hoping to buy their next property.

7. Conclusion

Through the use of modern machine learning using linear regression, it was concluded that the model provided a highly accurate **range** of values that a specific property would lie in. Most importantly, income was a major factor in predicting housing value which mirrors real life circumstances where more wealthy individuals tend to purchase more luxury residences in more affluent communities. One of the primary limitations to this project was the scope and size of data that was provided. Provided more additional features such as proximity to ocean, school quality, and crime rates, the model would be much more accurate in predicting its results and more precise estimation ranges. Furthermore, integrating more complex regression algorithms such as quadratic regression or Random Forests could capture more intricate relationships that linear regression is unable to fully account for. Moving forward, the next logical step would be to expand the given dataset by collecting more comprehensive housing data across California and experiment by manipulating different predictive variables and model architectures.

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